

Crosstalk and Noise

This chapter describes the signal integrity aspects of an ASIC in nanometer technologies. In deep submicron technologies, crosstalk plays an important role in the signal integrity of the design. The crosstalk noise refers to unintentional coupling of activity between two or more signals. Relevant noise and crosstalk analysis techniques, namely glitch analysis and crosstalk analysis, allow these effects to be included during static timing analysis and are described in this chapter. These techniques can be used to make the ASIC behave robustly.

6.1 Overview

Noise refers to undesired or unintentional effects affecting the proper operation of the chip. In nanometer technologies, the noise can impact in terms of functionality or in terms of timing of the devices.

Why noise and signal integrity?

There are several reasons why the noise plays an important role in the deep submicron technologies:

- *Increasing number of metal layers:* For example, a $0.25\mu\text{m}$ or $0.3\mu\text{m}$ process has four or five metal layers and it increases to ten or higher metal layers in the 65nm and 45nm process geometries. Figure 4-1 depicts the multiple layers of the metal interconnect.
- *Vertically dominant metal aspect ratio:* This means that the wires are thin and tall unlike the wide and thin in the earlier process geometries. Thus, a greater proportion of the capacitance is comprised of sidewall coupling capacitance which maps into wire to wire capacitance between neighboring wires.
- *Higher routing density due to finer geometry:* Thus, more metal wires are packed in close physical proximity.
- *Larger number of interacting devices and interconnects:* Thus, greater number of active standard cells and signal traces are packed in the same silicon area causing a lot more interactions.
- *Faster waveforms due to higher frequencies:* Fast edge rates cause more current spikes as well as greater coupling impact on the neighboring traces and cells.
- *Lower supply voltage:* The supply voltage reduction leaves little margin for noise.

In this chapter, we study the effect of crosstalk noise in particular. The crosstalk noise refers to unintentional coupling of activity between two or more signals. The crosstalk noise is caused by the capacitive coupling between neighboring signals on the die. This results in switching activity on a net to cause unintentional effects on the coupled signals. The affected sig-

nal is called the **victim**, and the affecting signals are termed as **aggressors**. Note that two coupled nets can affect each other, and often a net can be a victim as well as an aggressor.

Figure 6-1 shows an example of a few signal traces coupled together. The distributed RC extraction of the coupled interconnect is depicted along with several drivers and fanout cells. In this example, nets *N1* and *N2* have $Cc1 + Cc4$ as coupling capacitance between them, whereas $Cc2 + Cc5$ is the coupling capacitance between nets *N2* and *N3*.

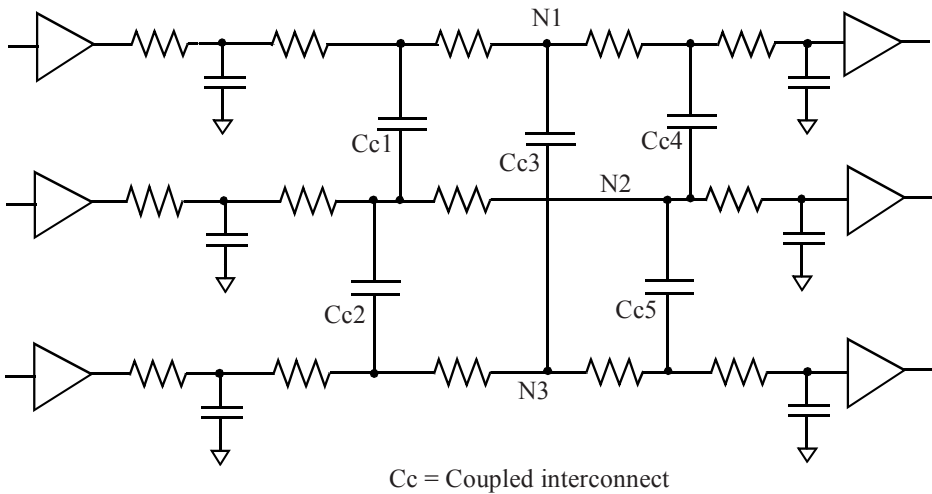


Figure 6-1 *Example of coupled interconnect.*

Broadly, there are two types of noise effects caused by crosstalk - **glitch**¹, which refers to noise caused on a steady victim signal due to coupling of switching activity of neighboring aggressors, and change in **timing** (crosstalk delta delay), caused by coupling of switching activity of the victim with the switching activity of the aggressors. These two types of crosstalk noise are described in the next two sections.

1. Some analysis tools refer to glitch as *noise*. Similarly, some tools use *crosstalk* to refer to crosstalk effect on delay.

6.2 Crosstalk Glitch Analysis

6.2.1 Basics

A steady signal net can have a glitch (positive or negative) due to charge transferred by the switching aggressors through the coupling capacitances. A positive glitch induced by crosstalk from a rising aggressor net is illustrated in Figure 6-2. The coupling capacitance between the two nets is depicted as one lumped capacitance C_c instead of distributed coupling, this is to simplify the explanation below without any loss of generality. In typical representations of the extracted netlist, the coupling capacitance may be distributed across multiple segments as seen previously in Section 6.1.

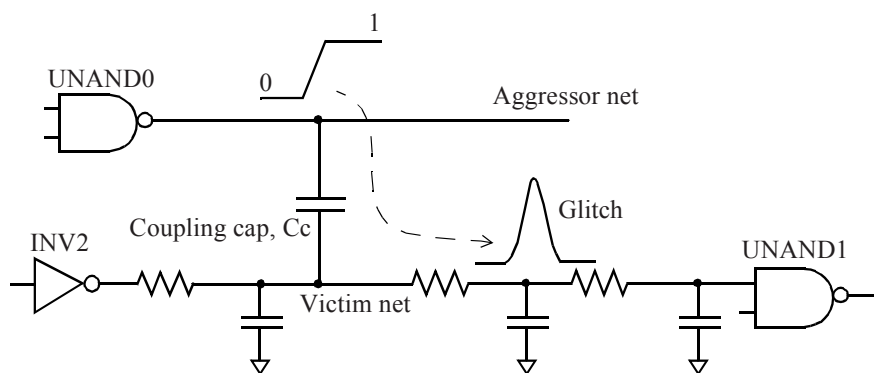


Figure 6-2 *Glitch due to an aggressor.*

In this example, the NAND2 cell UNAND0 switches and charges its output net (labeled *Aggressor*). Some of the charge is also transferred to the victim net through the coupling capacitance C_c and results in the positive glitch. The amount of charge transferred is directly related to the coupling capacitance, C_c , between the aggressor and the victim net. The charge transferred on the grounded capacitances of the victim net causes the glitch at that net. The steady value on the victim net (in this case, 0 or low) is restored because the transferred charge is dissipated through the pull-down stage of the driving cell INV2.

The magnitude of the glitch caused is dependent upon a variety of factors. Some of these factors are:

- i. *Coupling capacitance between the aggressor net and victim*: The greater the coupling capacitance, the larger the magnitude of the glitch.
- ii. *Slew of the aggressor net*: The faster the slew at the aggressor net, the larger the magnitude of glitch. In general, faster slew is because of higher output drive strength for the cell driving the aggressor net.
- iii. *Victim net grounded capacitance*: The smaller the grounded capacitance on the victim net, the larger the magnitude of the glitch.
- iv. *Victim net driving strength*: The smaller the output drive strength of the cell driving the victim net, the larger the magnitude of the glitch.

Overall, while the steady value on the victim net gets restored, the glitch can affect the functionality of the circuit for the reasons stated below.

- The glitch magnitude may be large enough to be seen as a different logic value by the fanout cells (e.g. a victim at logic-0 may appear as logic-1 for the fanout cells). This is especially critical for the sequential cells (flip-flops, latches) or memories, where a glitch on the clock or asynchronous set/reset can be catastrophic to the functionality of the design. Similarly, a glitch on the data signal at the latch input can cause incorrect data to be latched which can also be catastrophic if the glitch occurs when the data is being clocked in.
- Even if the victim net does not drive a sequential cell, a wide glitch may be propagated through the fanouts of the victim net and reach a sequential cell input with catastrophic consequences for the design.

6.2.2 Types of Glitches

There are different types of glitches.

Rise and Fall Glitches

The discussion in the previous subsection illustrates a positive or **rise glitch** on a victim net which is steady low. An analogous case is of a negative glitch on a steady high signal. A falling aggressor net induces a **fall glitch** on a steady high signal.

Overshoot and Undershoot Glitches

What happens when a rising aggressor couples to a victim net which is steady high? There is still a glitch which takes the victim net voltage above its steady high value. Such a glitch is called an **overshoot glitch**. Similarly, a falling aggressor when coupled to a steady low victim net causes an **undershoot glitch** on the victim net.

All four cases of glitches induced by crosstalk are illustrated in Figure 6-3.

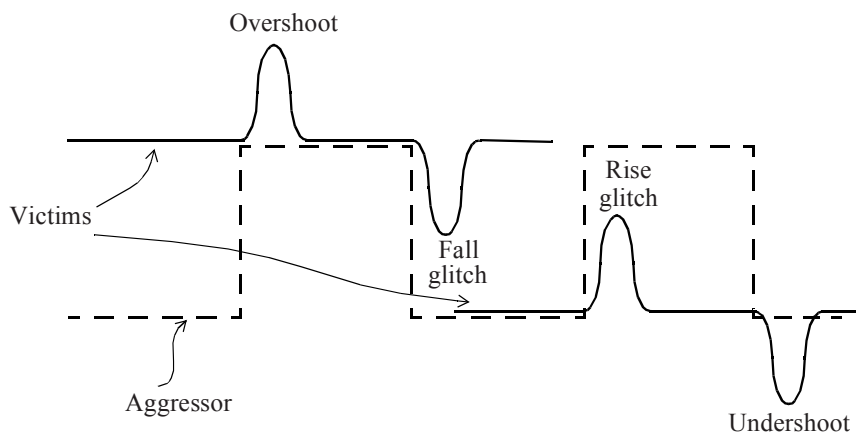


Figure 6-3 *Types of glitches.*

As described in the previous subsection, the glitch is governed by the coupling capacitance, aggressor slew and the drive strength of the victim net. The glitch computation is based upon the amount of current injected by the switching aggressor, the RC interconnect for the victim net, and the output impedance of the cell driving the victim net. The detailed glitch calculation is based upon the library models; the related noise models for the calculation are part of the standard cell library models described in Chapter 3. The output *dc_current* models in Section 3.7 relate to the output impedance of the cell.

6.2.3 Glitch Thresholds and Propagation

How can it be determined whether a glitch at a net can be propagated through the fanout cells? As discussed in an earlier subsection, a glitch caused by coupling from a switching aggressor can propagate through the fanout cell depending upon the fanout cell and glitch attributes such as glitch height and glitch width. This analysis can be based upon DC or AC noise thresholds. The DC noise analysis only examines the glitch magnitude and is conservative whereas the AC noise analysis examines other attributes such as glitch width and fanout cell output load. Various threshold metrics used in the DC and AC analyses of the glitches are described below.

DC Thresholds

The *DC noise margin* is a check used for glitch magnitude and refers to the DC noise limits on the input of a cell while ensuring proper logic functionality. For example, the output of an inverter cell may be high (that is, stay above the minimum value of *VOH*) as long as the input stays below the max value of *VIL* for the cell. Similarly, the output of the inverter cell may be low (that is, stay below *VOL* maximum value) as long as the input stays above the *VIH* minimum value. These limits are obtained based upon the DC transfer characteristics¹ of the cell and may be populated in the cell library.

1. See [DAL08] in Bibliography.

V_{OH} is the range of output voltage that is considered as a logic-one or high. V_{IL} is the range of input voltage that is considered a logic-zero or low. V_{IH} is the range of input voltage that is considered as a logic-one. V_{OL} is the range of output voltage that is considered as a logic-zero. An example of the input-output DC transfer characteristics of an inverter cell is given in Figure 6-4.

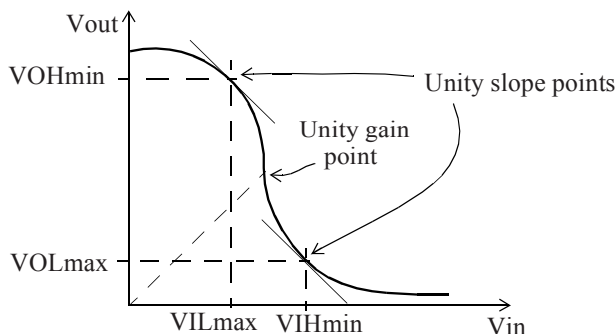


Figure 6-4 *DC transfer characteristics of an inverter cell.*

The V_{ILmax} and V_{IHmin} limits are also referred to as DC margin limits. The DC margins based upon V_{IH} and V_{IL} are steady state noise limits. These can thus be used as a filter for determining whether a glitch will propagate through the fanout cell. The DC noise margin limits are applicable for every input pin of a cell. In general, the DC margin limits are separate for *rise_glitch* (input low) and *fall_glitch* (input high). Models for DC margin can be specified as part of the cell library description. A glitch below the DC margin limit (for example, a rise glitch below the V_{ILmax} of the fanout pins) cannot be propagated through the fanout irrespective of the width of the glitch. Thus, a conservative glitch analysis checks that the peak voltage level (for all glitches) meets the V_{IL} and the V_{IH} levels of the fanout cells. As long as all nets meet the V_{IL} and V_{IH} levels for the fanout cells in spite of any glitches, it can be concluded that the glitches have no impact on the functionality of the design (since the glitches cannot cause the output to change).

Figure 6-5 shows an example of DC margin limits. The DC noise margin can also be fixed to the same limit for all nets in the design. One can set the largest tolerable noise (or glitch) magnitude, above which noise can be propagated through the cell to the output pin. Typically this check ensures that the glitch level is less than V_{ILmax} and greater than V_{IHmin} . The height is often expressed as a percent of the power supply. Thus, if the DC noise margin is set to 30%, that indicates that any glitch height greater than 30% of the voltage swing is identified as a potential glitch that can propagate through the cell and potentially impact the functionality of the design.

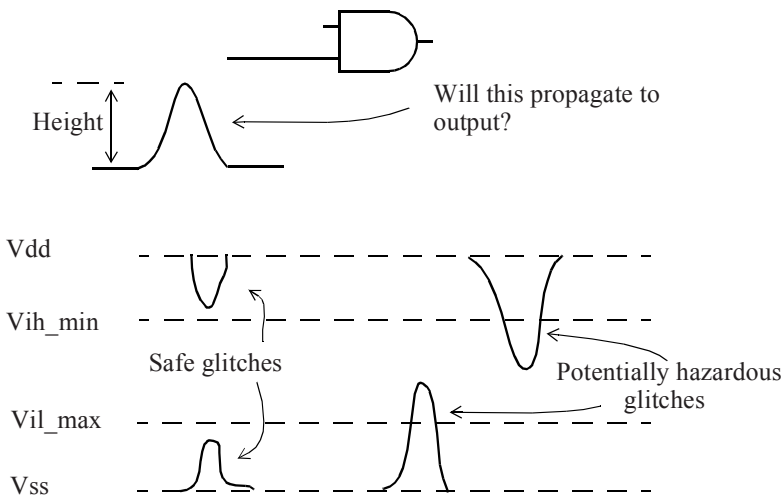


Figure 6-5 Glitch check based upon DC noise margin.

Not all glitches with magnitude larger than the DC noise margin can change the output of a cell. The width of the glitch is also an important consideration in determining whether the glitch will propagate to the output or not. A narrow glitch at a cell input will normally not cause any impact at the output of a cell. However, DC noise margin uses only a constant worst-case value irrespective of the signal noise width. See Figure 6-6. This provides a noise rejection level that is a very conservative estimate of the noise tolerance of a cell.

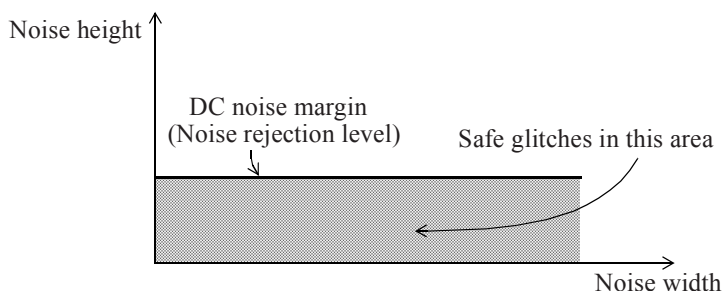


Figure 6-6 *DC noise rejection level.*

AC Thresholds

As described in the subsection above, the DC margin limits for glitch analysis are conservative since these analyze the design under the worst-case conditions. The DC margin limits verify that even if the glitch is arbitrarily wide, it will not affect the proper operation of the design.

In most cases, a design may not pass the conservative DC noise analysis limits. Therefore it becomes imperative to verify the impact of glitches with respect to the glitch width and the output load of the cell. In general, if the glitch is narrow or if the fanout cell has a large output capacitance, the glitch would have no effect on the proper functional operation. Both the effect of glitch width and the output capacitance can be explained in terms of the inertia of the fanout cell. In general, a single stage cell will stop any input glitch which is much narrower than the delay through the cell. This is because with a narrow glitch, the glitch is over before the fanout cell can respond to it. Thus, a very narrow glitch does not have any effect on the cell. Since the output load increases the delay through the cell, increasing the output load has the effect of minimizing the impact of glitch at the input - though it has the adverse effect of increasing the cell delay.

The AC noise rejection is illustrated in Figure 6-7 (for a fixed output capacitance). The dark shaded region represents *good* or *acceptable* glitches since these are either too narrow or too short, or both, and thus have no effect on the functional behavior of the cell. The lightly shaded region represents *bad*

or *unacceptable* glitches since these are too wide or too tall, or both, and thus such a glitch at the cell input affects the output of the cell. In the limiting case of very wide glitches, the glitch threshold corresponds to the DC noise margin as shown in Figure 6-7.

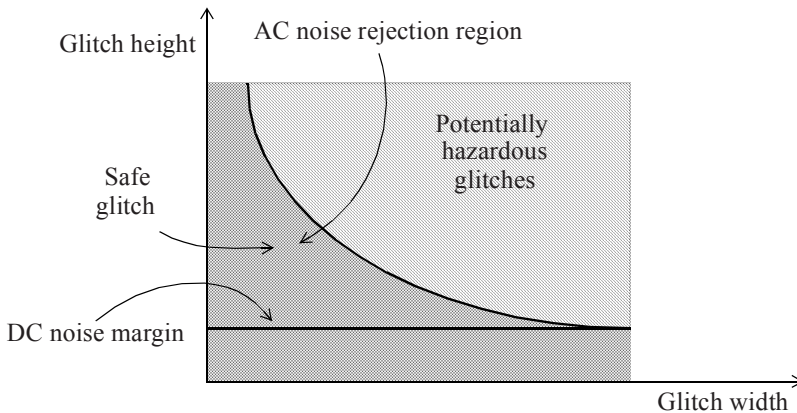
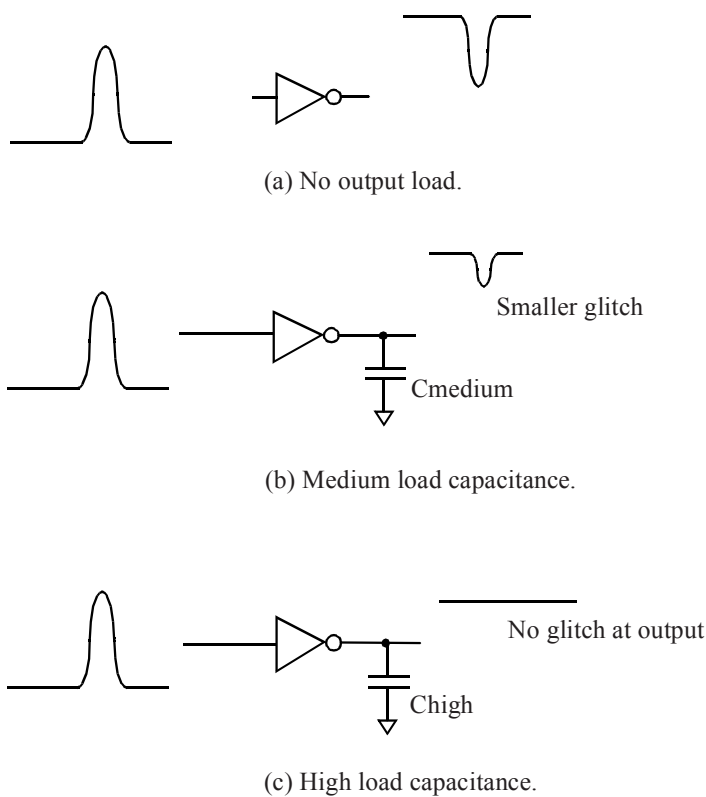


Figure 6-7 *AC noise rejection region.*

For a given cell, increasing the output load increases the noise margin since it increases the inertial delay and the width of the glitch that can pass through the cell. This phenomenon is illustrated through an example below. Figure 6-8(a) shows an unloaded inverter cell with a positive glitch at its input. The input glitch is taller than the DC margin of the cell and causes a glitch at the inverter output. Figure 6-8(b) shows the same inverter cell with some load at its output. The same input glitch at its input results in a much smaller glitch at the output. If the output load of the inverter cell is even higher, as in Figure 6-8(c), the output of the inverter cell does not have any glitch. Thus, increasing the load at the output makes the cell more immune to noise propagating from the input to the output.

As described above, the glitches below the AC threshold (the AC noise rejection region in Figure 6-7) can be ignored or the fanout cell can be considered to be immune from such a glitch. The AC threshold (or noise immunity) region depends upon the output load and the glitch width. As



* Input glitch size is same in all three cases.

Figure 6-8 *Output load determines size of propagated glitch.*

described in Chapter 3, noise immunity models include the effect of AC noise rejection described above. The *propagated_noise* models described in Section 3.7 capture the effect of AC noise threshold in addition to modeling the propagation through the cell.

What happens if the glitches are larger than the AC threshold? In such a case where the glitch magnitude exceeds the AC threshold, the glitch at the cell input produces another glitch at the output of the cell. The output glitch height and width is a function of input glitch width and height as well as the output load. This information is characterized in the cell library which contains detailed tables or functions for the output glitch magnitude and width as a function of the input pin glitch magnitude, glitch width and the load at the output pin. The glitch propagation is governed by *propagated_noise* models which are included in the library cell description. The *propagated_noise* (*low* and *high*) models are described in detail in Chapter 3.

Based upon the above, the glitch is computed at the output of the fanout cell and the same checks (and glitch propagation to the fanout) are followed at the fanout net and so on.

While we have used the generic term *glitch* in the discussion above, it should be noted that this applies separately to *rise glitch* (modeled by *propagated_noise_high*; or *noise_immunity_high* in earlier models), *fall glitch* (modeled by *propagated_noise_low*; or *noise_immunity_low* in earlier models), *overshoot glitch* (modeled by *noise_immunity_above_high*) and *undershoot glitch* (modeled by *noise_immunity_below_low*) as described in the previous sections.

To summarize, different inputs of a cell have different limits on the glitch threshold which is a function of glitch width and output capacitance. These limits are separate for input high (low transition glitch) and for input low (high transition glitch). The noise analysis examines the peak as well as the width of the glitch and analyzes whether it can be neglected or whether it can propagate to fanouts.

6.2.4 Noise Accumulation with Multiple Aggressors

Figure 6-9 depicts the coupling due to a single aggressor net switching and introducing a crosstalk glitch on the victim net. In general, a victim net may be capacitive-coupled to many nets. When multiple nets switch concurrently, the crosstalk coupling noise effect on the victim is compounded due to multiple aggressors.

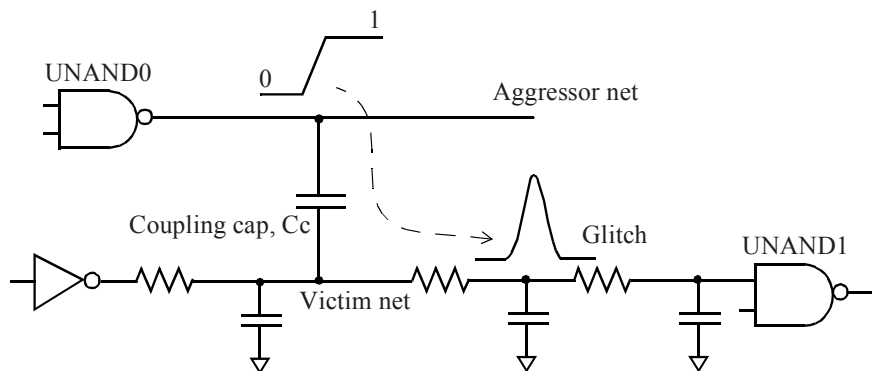


Figure 6-9 *Glitch from single aggressor.*

Most analyses for coupling due to multiple aggressors add the glitch effect due to each aggressor and compute the cumulative effect on the victim. This may appear conservative, however it does indicate the worst-case glitch on the victim. An alternate approach is the use of RMS (*root-mean-squared*) approach. When using the RMS option, the magnitude of the glitch on the victim is computed by taking the root-mean-square of the glitches caused by individual aggressors.

6.2.5 Aggressor Timing Correlation

For crosstalk glitch due to multiple aggressors, the analysis must include the timing correlation of the aggressor nets and determine whether the multiple aggressors can switch concurrently. The STA obtains this information from the timing windows of the aggressor nets. During timing

analysis, the *earliest* and the *latest* switching times of the nets are obtained. These times represent the *timing windows* during which a net may switch within a clock cycle. The switching windows (rising and falling) provide the necessary information on whether the aggressor nets can switch together.

Based upon whether the multiple aggressors can switch concurrently, the glitches due to individual aggressors are combined for the victim net. As a first step, the glitch analysis computes the four types of glitches (rise, fall, undershoot, and overshoot) separately for each potential aggressor. The next step combines the glitch contributions from the various individual aggressors. The multiple aggressors can combine separately for each type of glitch. For example, consider a victim net V coupled to aggressor nets $A1$, $A2$, $A3$ and $A4$. During analysis, it is possible that $A1$, $A2$, and $A4$ contribute to rising and overshoot glitches, whereas only $A2$ and $A3$ contribute to undershoot and falling glitches.

Consider another example where four aggressor nets can cause a rising glitch when the aggressor nets transition. Figure 6-10 shows the timing windows and the glitch magnitude caused by each aggressor net. Based upon the timing windows, the glitch analysis determines the worst possible combination of aggressor switching which results in the largest glitch. In this example, the switching window region is divided in four bins - each bin shows the possible aggressors switching. The glitch contribution from each aggressor is also depicted in Figure 6-10. Bin 1 has $A1$ and $A2$ switching which can result in a glitch magnitude of 0.21 ($= 0.11 + 0.10$). Bin 2 has $A1$, $A2$, and $A3$ switching which can result in a glitch magnitude of 0.30 ($= 0.11 + 0.10 + 0.09$). Bin 3 has $A1$ and $A3$ switching which can result in a glitch magnitude of 0.20 ($= 0.11 + 0.09$). Bin 4 has $A3$ and $A4$ switching which can result in a glitch magnitude of 0.32 ($= 0.09 + 0.23$).

Thus, bin 4 has the worst possible glitch magnitude of 0.32. Note that an analysis without using timing windows will predict a combined glitch magnitude of 0.53 ($= 0.11 + 0.10 + 0.09 + 0.23$) which can be overly pessimistic.

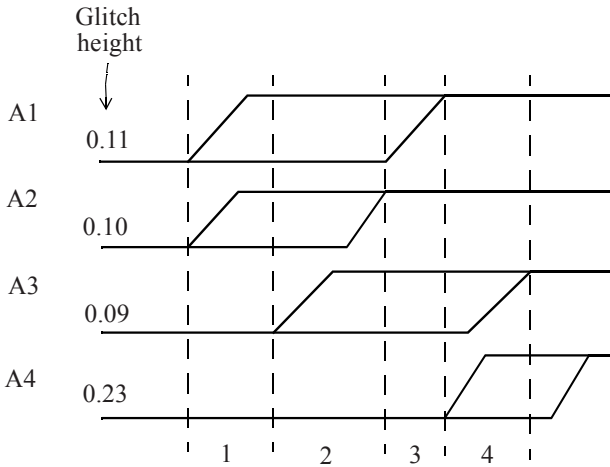


Figure 6-10 *Switching windows and glitch magnitudes from multiple aggressors.*

6.2.6 Aggressor Functional Correlation

For multiple aggressors, the use of timing windows reduces the pessimism in the analysis by considering the switching window during which a net can possibly switch. In addition, another factor to be considered is the *functional correlation* between various signals. For example, the scan¹ control signals only switch during the scan mode and are steady during functional or mission mode of the design. Thus, the scan control signals cannot cause a glitch on other signals during the functional mode. The scan control signals can only be aggressors during the scan mode. In some cases, the test and functional clocks are mutually exclusive such that the test clock is active only during testing when the functional clocks are turned off. In these designs, the logic controlled by test clocks and the logic controlled by functional clocks create two disjoint sets of aggressors. For such cases, the ag-

1. DFT test mode.

gressors controlled by test clocks cannot be combined with the other aggressors controlled by functional clocks for worst-case noise computation. Another example of functional correlation is two aggressors which are the complement (logical inverse) of each other. For such cases, both signal and its complement cannot be switching in the same direction for crosstalk noise computation.

Figure 6-11 shows an example of net $N1$ having coupling with three other nets $N2$, $N3$ and $N4$. In functional correlation, the functionality of the nets needs to be considered. Assume net $N4$ is a constant (for example, a mode setting net) and thus cannot be an aggressor on net $N1$, in spite of its coupling. Assume $N2$ is a net that is part of debug bus, but is in steady state in functional mode. Thus, net $N2$ cannot be an aggressor for net $N1$. Assuming net $N3$ carries functional data, only net $N3$ can be considered as a potential aggressor for net $N1$.

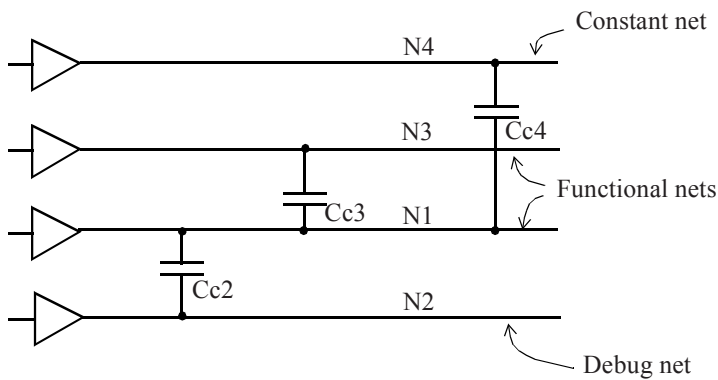


Figure 6-11 *Three couplings but only one aggressor.*

6.3 Crosstalk Delay Analysis

6.3.1 Basics

The capacitance extraction for a typical net in a nanometer design consists of contributions from many neighboring conductors. Some of these are grounded capacitances while many others are from traces which are part of other signal nets. The grounded as well as inter-signal capacitances are illustrated in Figure 6-1. All of these capacitances are considered as part of the total net capacitance during the basic delay calculation (without considering any crosstalk). When the neighboring nets are steady (or not switching), the inter-signal capacitances can be treated as grounded. When a neighboring net is switching, the charging current through the coupling capacitance impacts the timing of the net. The equivalent capacitance seen from a net can be larger or smaller based upon the direction of the aggressor net switching. This is explained in a simple example below.

Figure 6-12 shows net *N1* which has a coupling capacitance C_c to a neighboring net (labeled *Aggressor*) and a capacitance C_g to ground. This example assumes that the net *N1* has a rising transition at the output and considers different scenarios depending on whether or not the aggressor net is switching at the same time.

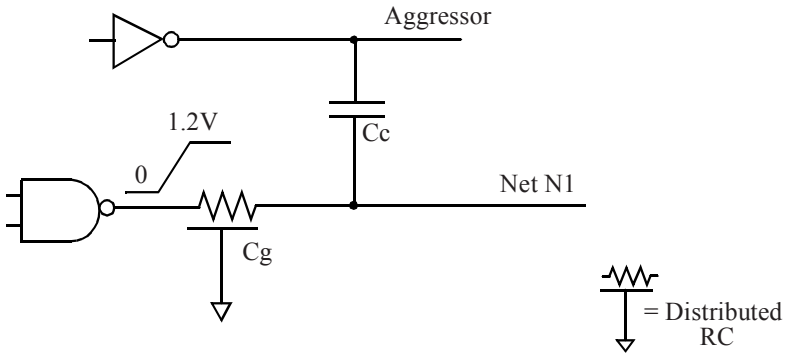


Figure 6-12 *Crosstalk impact example.*

The capacitive charge required from the driving cell in various scenarios can be different as described next.

- i. **Aggressor net steady.** In this scenario, the driving cell for the net $N1$ provides the charge for C_g and C_c to be charged to V_{dd} . The total charge provided by the driving cell of this net is thus $(C_g + C_c) * V_{dd}$. The base delay calculation obtains delays for this scenario where no crosstalk is considered from the aggressor nets. Table 6-13 shows the charge on C_g and C_c before and after the switching of $N1$ for this scenario.

<i>Capacitance</i>		<i>Before rising transition at net N1</i>	<i>After rising transition at net N1</i>
Grounded Cap, C_g		$V(C_g) = 0$	$V(C_g) = V_{dd}$
Coupling Cap, C_c	Aggressor net steady LOW	$V(C_c) = 0$	$V(C_c) = V_{dd}$
	Aggressor net steady HIGH	$V(C_c) = -V_{dd}$	$V(C_c) = 0$

Table 6-13 *Base delay calculation - no crosstalk.*

- ii. **Aggressor switching in same direction.** In this scenario, the driving cell is aided by the aggressor switching in the same direction. If the aggressor transitions at the same time with the same slew (identical transition time), the total charge provided by the driving cell is only $(C_g * V_{dd})$. If the slew of the aggressor net is faster than that of $N1$, the actual charge required can be even smaller than $(C_g * V_{dd})$ since the aggressor net can provide charging current for C_g also. Thus, the required charge from the driving cell with the aggressor switching in same direction is smaller than the corresponding charge for the steady aggressor described in Table 6-13. Therefore, the aggressor switching in the same direction results in a smaller delay for the switching net

N1; the reduction in delay is labeled as **negative crosstalk delay**. See Table 6-14. This scenario is normally considered for min path analysis.

<i>Capacitance</i>	<i>Before rising transition at net N1 and aggressor net</i>	<i>After rising transition at net N1 and aggressor net</i>
Grounded Cap, C_g	$V(C_g) = 0$	$V(C_g) = V_{dd}$
Coupling Cap, C_c	$V(C_c) = 0$	$V(C_c) = 0$

Table 6-14 *Aggressor switching in same direction - negative crosstalk.*

- iii. **Aggressor switching in opposite direction.** In this scenario, the coupling capacitance is charged from $-V_{dd}$ to V_{dd} . Thus, the charge on coupling capacitance changes by $(2 * C_c * V_{dd})$ before and after the transitions. This additional charge is provided by both the driving cell of net *N1* as well as the aggressor net. This scenario results in a larger delay for the switching net *N1*; the increase in delay is labeled as **positive crosstalk delay**. See Table 6-15. This scenario is normally considered for max path analysis.

<i>Capacitance</i>	<i>Before transition at net N1 and aggressor net (net N1 is low; aggressor net is high)</i>	<i>After transition (net N1 is high; and aggressor net is low)</i>
Grounded Cap, C_g	$V(C_g) = 0$	$V(C_g) = V_{dd}$
Coupling Cap, C_c	$V(C_c) = -V_{dd}$	$V(C_c) = V_{dd}$

Table 6-15 *Aggressor switching in opposite direction - positive crosstalk.*

The above example illustrates the charging of C_c in various cases and how it can impact the delay of the switching net (labelled as $N1$). The example considers only a rising transition at net $N1$, however similar analysis holds for falling transitions also.

6.3.2 Positive and Negative Crosstalk

The base delay calculation (without any crosstalk) assumes that the driving cell provides all the necessary charge for rail-to-rail transition of the total capacitance of a net, C_{total} ($= C_{ground} + C_c$). As described in the previous subsection, the charge required for the coupling capacitance C_c is larger when the coupled (aggressor) net and victim net are switching in the opposite directions. The aggressor switching in the opposite direction increases the amount of charge required from the driving cell of the victim net and increases the delays for the driving cell and the interconnect for the victim net.

Similarly, when the coupled (aggressor) net and the victim net are switching in the same direction, the charge on C_c remains the same before and after the transitions of the victim and aggressor. This reduces the charge required from the driving cell of the victim net. The delays for the driving cell and the interconnect for the victim net are reduced.

As described above, concurrent switching of victim and aggressor affects the timing of the victim transition. Depending upon the switching direction of the aggressor, the crosstalk delay effect can be positive (slow down the victim transition) or negative (speed up the victim transition).

An example of positive crosstalk delay effect is shown in Figure 6-16. The aggressor net is rising at the same time when the victim net has a falling transition. The aggressor net switching in opposite direction increases the delay for the victim net. The positive crosstalk impacts the driving cell as well as the interconnect - the delay for both of these gets increased.

The case of negative crosstalk delay is illustrated in Figure 6-17. The aggressor net is rising at the same time as the victim net. The aggressor net switching in the same direction as the victim reduces the delay of the vic-

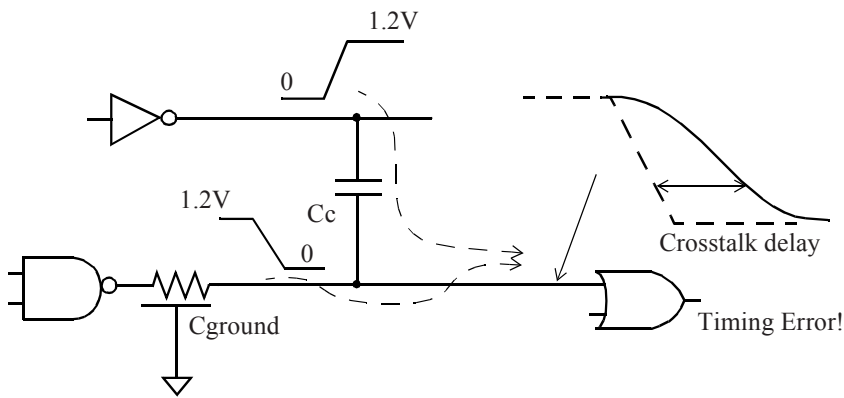


Figure 6-16 *Positive crosstalk delay.*

tim net. As before, the negative crosstalk affects the timing of the driving cell as well as the interconnect - the delay for both of these is reduced.

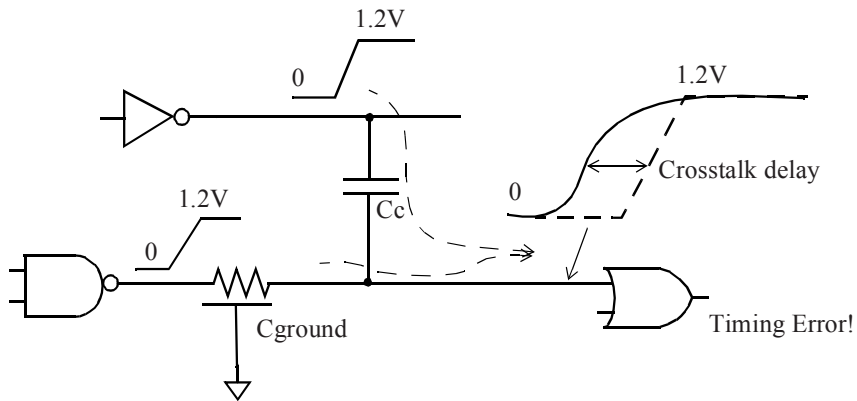


Figure 6-17 *Negative crosstalk delay.*

Note that the worst positive and worst negative crosstalk delays are computed separately for rise and fall delays. The worst set of aggressors for the

rise max, rise min, fall max, fall min delays with crosstalk are, in general, different. This is described in the subsections below.

6.3.3 Accumulation with Multiple Aggressors

The crosstalk delay analysis with multiple aggressors involves accumulating the contributions due to crosstalk for each of the aggressors. This is similar to the analysis for crosstalk glitches described in Section 6.2. When multiple nets switch concurrently, the crosstalk delay effect on the victim gets compounded due to multiple aggressors.

Most analyses for coupling due to multiple aggressors add the incremental contribution from each aggressor and compute the cumulative effect on the victim. This may appear conservative, however it does indicate the worst-case crosstalk delay on the victim.

Similar to the analysis of multiple aggressors for crosstalk glitch analysis, contributions can also be added using root-mean-squared (RMS) which is less pessimistic than the straight sum of individual contributions.

6.3.4 Aggressor Victim Timing Correlation

The handling of timing correlation for crosstalk delay analysis is conceptually similar to the timing correlation for crosstalk glitch analysis described in Section 6.2. The crosstalk can affect the delay of the victim, only if the aggressor can switch at the same time as the victim. This is determined using the timing windows of the aggressor and the victim. As described in Section 6.2, the *timing windows* represent the *earliest* and the *latest* switching times during which a net may switch within a clock cycle. If the timing windows of the aggressor and the victim overlap, the crosstalk effect on delay is computed. For multiple aggressors, the timing windows for multiple aggressors are also analyzed similarly. The possible effect in various timing bins is computed and the timing bin with the worst crosstalk delay impact is considered for delay analysis.

Consider the example below where three aggressor nets can impact the timing of the victim net. The aggressor nets (A1, A2, A3) are capacitively coupled to the victim net (V) and also their timing windows overlap with that of the victim. Figure 6-18 shows the timing windows and the possible crosstalk delay impact caused by each aggressor. Based upon the timing windows, the crosstalk delay analysis determines the worst possible combination of the aggressor switching that causes the largest crosstalk delay impact. In this example, the timing window overlap region is divided into three bins - each bin shows the possible aggressors switching. Bin 1 has A1 and A2 switching which can result in crosstalk delay impact of 0.26 ($= 0.12 + 0.14$). Bin 2 has A1 switching which can result in crosstalk delay impact of 0.14. Bin 3 has A3 switching which can result in crosstalk delay impact of 0.23. Thus, bin 1 has the worst possible crosstalk delay impact of 0.26.

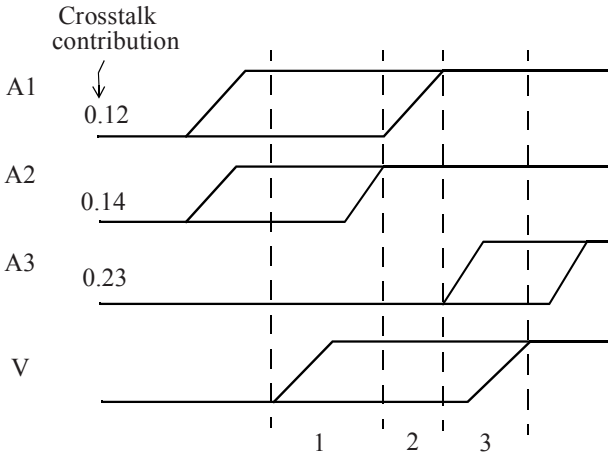


Figure 6-18 *Timing windows and crosstalk contributions from various aggressors.*

As indicated previously, the crosstalk delay analysis computes the four types of crosstalk delays separately. The four types of crosstalk delays are **positive rise delay** (rise edge moves forward in time), **negative rise delay** (rise edge moves backward in time), **positive fall delay** and **negative fall delay**. In general, a net can have different sets of aggressors in each of these

four cases. For example, a net can be coupled to aggressors $A1$, $A2$, $A3$ and $A4$. During crosstalk delay analysis, it is possible that $A1$, $A2$, $A4$ contribute to positive rise and negative fall delay contributions whereas $A2$ and $A3$ contribute to negative rise and positive fall delay contributions.

6.3.5 Aggressor Victim Functional Correlation

In addition to timing windows, crosstalk delay calculation can consider the *functional correlation* between various signals. For example, the scan control signals only switch during the scan mode and are steady during functional or mission mode of the design. Thus, the scan control signals can-not be aggressors during the functional mode. The scan control signals can only be aggressors during the scan mode in which case these signals can-not be combined with the other functional signals for worst-case noise computation.

Another example of functional correlation is a scenario where two aggressors are complements of each other. For such cases, both signal and its complement can never be switching in the same direction for crosstalk noise computation. This type of functional correlation information, when available, can be utilized so that the crosstalk analysis results are not pessimistic by ensuring that only the signals which can actually switch together are included as aggressors.

6.4 Timing Verification Using Crosstalk Delay

The following four types of crosstalk delay contributions are computed for every cell and interconnect in the design:

- i.* Positive rise delay (rise edge moves forward in time)
- ii.* Negative rise delay (rise edge moves backward in time)
- iii.* Positive fall delay (fall edge moves forward in time)
- iv.* Negative fall delay (fall edge moves backward in time)

The crosstalk delay contributions are then utilized during timing analysis for the verification of the max and min paths (setup and hold checks). The clock path for the launch and capture flip-flops are handled differently. The details of the data path and clock path analyses for the setup and hold checks are described in this section.

6.4.1 Setup Analysis

The STA with crosstalk analysis verifies the design with the worst-case crosstalk delays for the data path and the clock paths. Consider the logic shown in Figure 6-19 where crosstalk can occur at various nets along the data path and along the clock paths. The worst condition for setup check is when both the launch clock path and the data path have positive crosstalk and the capture clock path has negative crosstalk. The positive crosstalk contributions on launch clock path and data path delay the arrival of data at the capture flip-flop. In addition, the negative crosstalk on capture clock path results in capture flip-flop being clocked early.

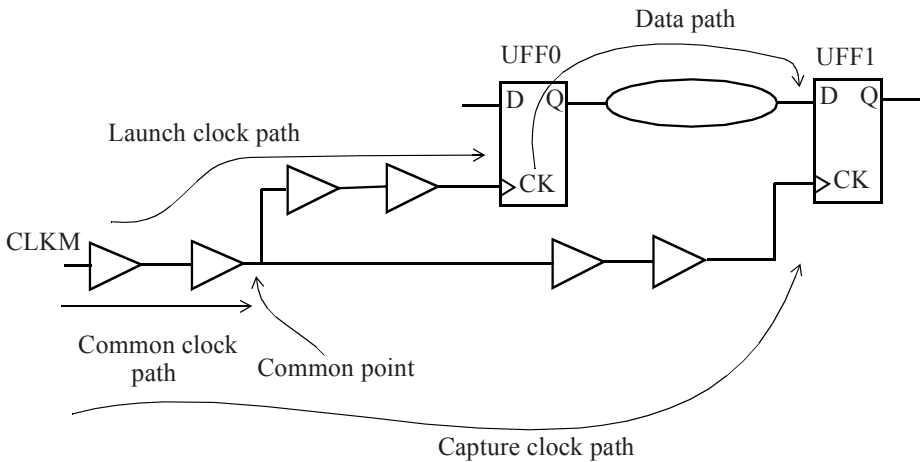


Figure 6-19 *Crosstalk for data path and clock path.*

Based upon above description, the setup (or max path) analysis assumes that:

- Launch clock path sees positive crosstalk delay so that the data is launched late.
- Data path sees positive crosstalk delay so that it takes longer for the data to reach the destination.
- Capture clock path sees negative crosstalk delay so that the data is captured by the capture flip-flop early.

Since the launch and capture clock edges for a setup check are different (normally one clock cycle apart), the common clock path (Figure 6-19) can have different crosstalk contributions for the launch and capture clock edges.

6.4.2 Hold Analysis

The worst-case hold (or min path) analysis for STA is analogous to the worst-case setup analysis described in the preceding subsection. Based upon the logic shown in Figure 6-19, the worst condition for hold check occurs when both the launch clock path and the data path have negative crosstalk and the capture clock path has positive crosstalk. The negative crosstalk contributions on launch clock path and data path result in early arrival of the data at the capture flip-flop. In addition, the positive crosstalk on capture clock path results in capture flip-flop being clocked late.

There is one important difference between the hold and setup analyses related to crosstalk on the common portion of the clock path. The launch and capture clock edge are normally the same edge for the hold analysis. The clock edge through the common clock portion cannot have different crosstalk contributions for the launch clock path and the capture clock path. Therefore, the worst-case hold analysis removes the crosstalk contribution from the common clock path.

The worst-case hold (or min path) analysis for STA with crosstalk assumes:

- Launch clock (not including the common path) sees negative crosstalk delay so that the data is launched early.
- Data path sees negative crosstalk delay so that it reaches the destination early.
- Capture clock (not including the common path) sees positive crosstalk delay so that the data is captured by the capture flip-flop late.

As described above, the crosstalk impact on the common portion of the clock tree is *not* considered for the hold analysis. The positive crosstalk contribution of the launch clock and negative crosstalk contribution of the capture clock are only computed for the *non-common* portions of the clock tree. In STA reports for hold analysis, the common clock path may show different crosstalk contributions for the launch clock path and the capture clock path. However, the crosstalk contributions from the common clock path are removed as a separate line item labeled as common path pessimism removal. Examples of common path pessimism removal in STA reports are provided in Section 10.1.

As described in the preceding subsection, the setup analysis concerns two different edges of the clock which may potentially be impacted differently in time. Thus, the common path crosstalk contributions are considered for both the launch and the capture clock paths during setup analysis.

The clock signals are critical since any crosstalk on the clock tree directly translates into clock jitter and impacts the performance of the design. Thus, special considerations should be adopted for reducing crosstalk on the clock signals. A common noise avoidance method is shielding of the clock tree - this is discussed in detail in Section 6.6.

6.5 Computational Complexity

A large nanometer design is generally too complex to allow for every coupling capacitance to be analyzed with reasonable turnaround time. The parasitic extraction of a typical net contains coupling capacitances to many neighboring signals. A large design will normally require appropriate settings for the parasitic extraction and crosstalk delay and glitch analyses. These settings are selected to provide acceptable accuracy for the analyses while ensuring that the CPU requirements remain feasible. This section describes some of the techniques that can be used for the analysis of a large nanometer design.

Hierarchical Design and Analysis

Hierarchical methodology for verifying a large design was introduced in Section 4.5. A similar approach is also applicable for reducing the complexity of extraction and analyses.

For a large design, it is normally not practical to obtain parasitic extraction in one run. The parasitics for each hierarchical block can be extracted separately. This in turn requires that a hierarchical design methodology be used for the design implementation. This implies that there be no coupling between signals inside the hierarchical block and signals outside the block. This can be achieved either with no routing over the block or by adding a shield layer over the block. In addition, signal nets should not be routed close to the boundary of the block and any nets routed close to the boundary of the block should be shielded. This avoids any coupling with the nets from other blocks.

Filtering of Coupling Capacitances

Even for a medium sized block, the parasitics will normally include a large number of very small coupling capacitances. The small coupling capacitances can be filtered during extraction or during the analysis procedures.

This filtering can be based upon the following criteria:

- i. *Small value*: Very small coupling capacitances, for example, below 1fF, can be ignored for the crosstalk or noise analysis. During extraction, the small couplings can be treated as grounded capacitances.
- ii. *Coupling ratio*: The impact of coupling on a victim is based upon the relative value of the coupling capacitance to the total capacitance of the victim net. Aggressor nets with a small coupling ratio, for example, below 0.001, can be excluded from crosstalk delay or glitch analyses.
- iii. *Lumping small aggressors together*: Multiple aggressors with very small contributions can be mapped to one larger virtual aggressor. This can be pessimistic but can simplify the analysis. Some of the possible pessimism can be mitigated by switching a subset of the aggressors. The exact subset of switching aggressors can be determined by statistical methods.

6.6 Noise Avoidance Techniques

The preceding sections described the impact and analysis of crosstalk effects. In this section, we describe some noise avoidance techniques which can be utilized in the physical design phase.

- i. *Shielding*: This method requires that shield wires are placed on either side of the critical signals. The shields are connected to power or ground rails. The shielding of critical signals ensures that there are no active aggressors for the critical signals since the nearest neighbors in the same metal layer are shield traces at a fixed potential. While there can be some coupling from routes in the different metal layers, most of the coupling capacitances are due to the capacitive coupling in the same layer. Since the immediate metal layers (above and below) would normally be routed orthogonally, the capacitive coupling across layers is

minimized. Thus, placing shield wires in the same metal layer ensures that there is minimal coupling for the critical signals. In cases where shielding with ground or power rails is not possible due to routing congestion, signal with low switching activity such as scan control which are fixed during functional mode can be routed as immediate neighbors for the critical signals. These shielding approaches ensure that there is no crosstalk due to capacitive coupling of the neighbors.

- ii. *Wire spacing*: This reduces the coupling to the neighboring nets.
- iii. *Fast slew rate*: A fast slew rate on the net implies that the net is less susceptible to crosstalk and is inherently immune to cross-talk effects.
- iv. *Maintain good stable supply*: This is important not for crosstalk but for minimizing jitter due to power supply variations. Significant noise can be introduced on the clock signals due to noise on the power supply. Adequate decoupling capacitances should be added to minimize noise on the power supply.
- v. *Guard ring*: A guard ring (or double guard ring) in the substrate helps in shielding the critical analog circuitry from digital noise.
- vi. *Deep n-well*: This is similar to the above as having deep n-well¹ for the analog portions helps prevent noise from coupling to the digital portions.
- vii. *Isolating a block*: In a hierarchical design flow, routing halos can be added to the boundary of the blocks; furthermore, isolation buffers could be added to each of the IO of the block.

□

1. See [MUK86] in Bibliography.